

Rahul Deb

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LinkedIn — GitHub — Portfolio

EXPERIENCE

UnitedHealth Group (Optum)

2022 – Present

Backend Engineer / Team Lead

- Architect and maintain scalable **Java Spring Boot microservices** powering enterprise healthcare platforms handling high-volume backend workloads.
- Improved API latency by **40%** by introducing asynchronous processing, concurrency optimization, and efficient resource management across backend services.
- Reduced cloud infrastructure costs by **35%** by optimizing Azure resource utilization and backend service architecture.
- Designed and implemented **Kafka-based event streaming pipelines** enabling reliable, fault-tolerant communication across distributed microservices.
- Led backend modernization initiatives including migration of legacy services to **Azure cloud infrastructure** using Terraform and GitHub Actions.
- Mentored engineers and led backend development efforts for a team of 2 developers, improving debugging practices and system reliability.
- Implemented **event-driven architecture** enabling scalable asynchronous communication across services.

ENGINEERING PROJECTS

Java 11 → Java 21 Migration

- Migrated **15+ production microservices** to Java 21 leveraging Virtual Threads for improved concurrency.
- Achieved **45% higher throughput** and **50% reduction in memory usage**.

Async Logging System (LMAX Disruptor)

- Implemented asynchronous logging pipeline using **Log4j2 and LMAX Disruptor**.
- Reduced request latency by **40%** in high-throughput backend systems.

Kafka Streaming Audit Platform

- Developed Kafka producers and consumers with **DLQ handling** for reliable event processing.
- Built monitoring dashboards for auditing events across distributed backend services.

Workflow Parallelization

- Refactored legacy backend workflows using asynchronous execution and custom thread pools.
- Reduced workflow latency from **2000ms to 1100ms**.

TECHNICAL SKILLS

Languages: Java, SQL, Bash

Backend & Architecture: Spring Boot, REST APIs, Microservices, Event-Driven Architecture, Distributed Systems

Streaming & Messaging: Apache Kafka, Message Queues, Asynchronous Processing, Fault-Tolerant Systems

Cloud & Infrastructure: Azure, Terraform, GitHub Actions, Infrastructure as Code

Observability & Performance: Application Insights, KQL Queries, JMeter, Performance Optimization

System Design Focus: Scalable Backend Systems, High-Throughput Services, Concurrent Processing

EDUCATION

National Institute of Technology, Silchar

Bachelor of Technology — Electronics and Communication Engineering